What makes motion hard to stop?

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare momentum changes in a controlled stopping situation.

SUCCESS CRITERIA - I can...

1. Compare mass and velocity while controlling other variables.
2. Use momentum change to compare stopping force for equal times.
3. Avoid claiming that momentum alone determines damage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “momentum” and “impulse” to compare your model or result. Sentence starters are optional.

momentum: mass times velocity, including direction

impulse: net force acting over time; change in momentum

system: the object or objects chosen for analysis

01 Warm-up: bring back an earlier idea

Two ideal Sun-orbits have $a_B = 3\ a_A$.
Find T_B/T_A.
Explain why the same normalized Sun-orbit equation is not automatically valid around another body.

Word help: orbit = a path around another body under gravity

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $p = mv$; $F_{avg} = \Delta p/\Delta t$.
Momentum p includes direction. For a fixed stopping time, greater momentum change needs greater average net force.

Worked reasoning: A 2 kg cart and a 4 kg cart move right at 3 m/s. Their momenta mv are 6 and 12 kg·m/s. Stopping the second requires twice the momentum change.

Your turn: Draw the velocity and momentum arrows for both carts.
Explain which pair has equal lengths.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Carts A and B both move right at 2 m/s. A has mass 1 kg; B has mass 3 kg. Which has more momentum?
State the controlled quantity.

Sentence starter: I notice … . This supports … because … .



A: 1 kg; 2 m/s B: 3 kg; 2 m/s

Marzano check - circle: ① ② ③

04 Apply the physics

Three carts undergo separate bumper tests. Cart A is 1 kg at +2 m/s; B is 3 kg at +2 m/s; C is 1 kg at +4 m/s. Each stops in 0.50 s.
Find the signed momentum change and average force in each trial.

Sentence starter: I found … [unit]. This value means … in this situation.

Cart	Mass (kg)	Initial v (m/s)	Final v
A	1	2	0
B	3	2	0
C	1	4	0

Marzano check - circle: ① ② ③

05 Explain your reasoning

Two objects have equal momentum but strike different materials over different times. Can equal momentum alone establish equal damage?
Explain 0.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

Two equal 4000 kg masses are 6 m apart.
Use $G=6.67\times10^{-11}$ N·m²/kg².
Find force. By what factor is an answer wrong if r rather than r^2 is used?
Label the masses, field or force relevant to the question.

Sentence starter: The step I would change is … . The evidence is … .



Label separation if needed.

Source Object

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A 3 kg cart travels right at 4 m/s and stops in 4 s. Take right as positive.
Find the signed impulse and average force.
Draw arrows for the initial momentum and stopping force.

Sentence starter: I found … [unit]. This value means … in this situation.

Cart stopping

Right is positive.



Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 0.5 kg cart at +8 m/s and a 2 kg cart at +2 m/s each stop in 0.2 s.
Compare their momentum changes and average stopping forces.

Sentence starter: Both … . They differ in … because … .

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: 1 2 3